DSE Bullet Series – Hardware Specifications and Upgrades (Solutions 01)

Name: _____ ( ) Class:____ Date: _____

1. Past Questions (2012–2023) #Learn from past mistakes

Year	Past Question	Focus / Marks
2012	Two computer specification tables; give three arguments	Match each specification to a benefit
2016	Cache memory and bus in a server	Reduce CPU waiting; transfer data/addresses
2019	RAM upgrade does not improve performance; what next?	CPU bottleneck; extra RAM only reduces paging
2021	Two differences: RAM vs SSD	Volatile/non-volatile; speed/capacity
2022	Quad-core vs dual-core; video editing needs?	Parallel processing; GPU, RAM, SSD
2023	HDD ➔ SSD; how RAM improves performance	No moving parts, faster; fewer SSD reads

2. Key Notes #Match specifications to benefits

CPU: three specifications — cores, clock rate and cache. More cores process more tasks at once; a higher clock rate (GHz) means more cycles per second; cache is a high-speed buffer that keeps recently used instructions and reduces CPU waiting. Bus width = bits transferred at one time.

RAM is volatile: cleared on power-off; temporarily holds running programs and data. More RAM loads more programs at once and reduces repeated SSD reads (paging), so switching windows is smoother.
*RAM does not speed up the CPU: a calculation-heavy program is limited by CPU speed. Add RAM only if RAM is insufficient; otherwise upgrade the CPU (more cores / higher clock rate).

ROM is non-volatile: keeps data on power-off; stores BIOS/firmware, is used first at boot and is normally not freely rewritable.

SSD vs HDD: an SSD has no moving parts ➔ shock-resistant, quiet and low power; shorter access time and higher data rate ➔ faster boot-up and loading. Drawbacks: higher cost per GB and limited write cycles.

Use C-F-B: Context (C) + Feature (F) + Benefit (B). Example: ‘Mei edits videos (C). A quad-core CPU processes calculations in parallel (F). Rendering time is shorter (B).’
*Common mistakes: writing only ‘fast’ or ‘good’; comparing without a common criterion; treating RAM as permanent storage.

3. Similar Questions #Same angle, different object

1. Mei compares A (8-core 2.4 GHz, 8 GB RAM, 256 GB) and B (4-core 2.0 GHz, 4 GB RAM, 128 GB).

(a) Use the specifications above to support the argument that A is more suitable for gaming.

【Answer】A has an octa-core CPU: more cores can process graphics and physics calculations in parallel, giving smoother gameplay; 8 GB RAM: can keep the game and background apps loaded, reducing lag; 256 GB storage: can install several large games. (Any two items + corresponding benefit)

(b) State **two** other hardware specifications to consider, and explain their purposes.

【Answer】Screen resolution/refresh rate: sharper images and smoother motion; battery capacity: longer usage time; weight: easier to carry; camera resolution: better photo quality. (Any two + purpose)

2. Ka-fai has added 8 GB RAM to his computer, but a program that performs many calculations is still slow.

(a) Explain why adding RAM cannot improve the performance of this program.

【Answer】The bottleneck is the CPU's processing speed. Adding RAM only increases temporary storage for programs and data; it does not increase the number of calculations the CPU can perform per second. If existing RAM is already sufficient, adding more RAM will not speed it up.

(b) Suggest one hardware component that should be upgraded and explain why.

【Answer】CPU: more cores can process calculations in parallel, or a higher clock rate increases the number of calculations per second, directly speeding up processing. (GPU is also acceptable if the program supports parallel processing.)

3. Ka-ching has taken many photos with a digital camera and needs to back them up.

(a) Compare a memory card and an external hard disk as backup media in **two** ways.

【Answer】Memory card: small and removable, but higher cost per GB and easier to lose; external hard disk: larger capacity and lower cost per GB, but has moving parts and is more vulnerable to shock. (Use a common criterion.)

(b) Explain why a camera uses a memory card instead of an internal hard disk when taking photos.

【Answer】A memory card is small, has no moving parts (shock-resistant) and can be inserted directly into the camera; an internal hard disk would increase the camera's size and power consumption, and is more vulnerable to shock.

DSE Bullet Series – Storage Devices and Backup (Solutions 02)

1. Past Questions (2013–2023) #HDD, SSD, flash — don't mix them up

Year	Past Question	Focus / Marks
2013	HDD ➔ SSD; state two benefits	No moving parts; faster; less power; quieter
2015	Tablet uses flash memory, not an HDD	Shock-resistant; less power; smaller; silent
2016	Function of a server's hard disk	Long-term storage of program and user data
2017	Flash memory vs RAM/ROM	Non-volatile vs volatile; rewritable vs read-only
2021	Two differences: RAM vs SSD	Volatile/non-volatile; speed/capacity
2023	HDD ➔ SSD; two benefits	No moving parts, shock-resistant; faster boot-up

2. Key Notes #Fast, expensive, small ➔ slow, cheap, large

Register ➔ cache ➔ RAM ➔ SSD/HDD ➔ optical disc ➔ magnetic tape. Upward: faster, costlier per GB, smaller; downward: slower, cheaper, larger.

Secondary storage is non-volatile: data remains after power-off; the CPU must first load data into RAM before processing.

An HDD uses rotating disks and read/write heads: large capacity and low cost per GB, but shock-sensitive, noisy and uses more power. An SSD uses flash chips and has no moving parts: short access time, shock-resistant, quiet and low power, but costs more per GB and has limited write cycles.

Flash memory (EEPROM) is non-volatile, electrically erasable and rewritable in blocks; used in USB drives, memory cards and SSDs.

An optical disc allows direct access: suitable for distribution and backup. Magnetic tape uses sequential access, with large capacity and low cost per GB: suitable for bulk archival backup.

When choosing backup media, consider capacity, speed, cost, durability, portability and whether it is kept offline/off-site.

*Common mistakes: treating RAM as permanent storage; comparing without a common criterion; writing only ‘fast’ or ‘large capacity’ without explaining the effect.

3. Similar Questions #Same angle, different medium

1. The school needs to back up ten years of event recordings. The files are large but are rarely viewed.

(a) Compare an optical disc and magnetic tape as backup media in **two** ways.

【Answer】Magnetic tape has larger capacity and lower cost per GB, suitable for bulk archiving; tape uses sequential access, so retrieving an individual file is slower, while an optical disc allows direct access. (Use a common criterion.)

(b) Apart from capacity, state one factor to consider when choosing backup media and explain it.

【Answer】Durability: tape/optical discs are affected by humidity and high temperature, which affects storage life; portability: removable media can be stored at another location, reducing fire and flood risk; cost: capacity available within the same budget.

2. The library computer room is replacing its internal storage. An SSD costs more per GB; an HDD is cheaper.

(a) Which one would you recommend? Give **two** reasons.

【Answer】SSD: faster boot-up and software loading (shorter access time); no moving parts, more shock-resistant, quieter and lower power, suitable for students who switch computers on and off frequently. (HDD is also acceptable: lower cost per GB and larger capacity for the same budget.)

(b) State one disadvantage of an SSD.

【Answer】Higher cost per GB; limited write cycles, so frequent heavy writing causes faster wear.

3. Ka-fai says: 'I have saved the files on an external hard disk, so I do not need a backup.'

(a) State **two** backup practices and explain the benefit of each.

【Answer】Offline backup: disconnect the media after backup to prevent ransomware from encrypting the backup; off-site backup: keep a copy at another location or in the cloud to reduce fire, flood and theft risk; regular backup: allows restoration of an earlier version.

(b) Explain why storing files on only one external hard disk is risky.

【Answer】Hardware may fail, be lost or stolen; ransomware may encrypt the connected hard disk; a single copy has no other version for recovery.

DSE Bullet Series – Input and Output Devices (Solutions 03)

1. Past Questions (2012–2021) #Input or output — don't mix them up

Year	Past Question	Focus / Marks
2012	Compare input and display device specifications of two computers	Keyboard vs virtual keyboard; LCD vs LED trade-off
2015	Input methods on a tablet other than a virtual keyboard	Microphone (voice), stylus, camera (scanning/QR)
2018	Touch screen vs keyboard and mouse; suggest an additional output device	Use a common criterion; explain the device's purpose
2020	Two considerations for lighting and curtains in a clinic	Ergonomics: avoid glare, even brightness
2021	Tablet specifications: physical keyboard vs lighter body	Portability vs typing efficiency trade-off

2. Key Notes #In, out, store — three actions

Input devices send data into the computer: keyboard, mouse/touchpad, scanner, microphone, digital camera and sensor. Output devices send results out: monitor, printer, projector and speaker.

Input devices are of two types: general (common on personal computers) and special (for specific commercial or industrial uses, e.g. OMR, barcode reader, magnetic stripe reader, RFID and sensor).

When choosing a device, consider purpose, speed, accuracy, cost, comfort/ergonomics and compatibility with existing systems. Comparisons must use a common criterion, e.g. ‘Compared with a keyboard, a touch screen is slower for entering text.’

Common specifications: scanner — resolution (dpi), colour depth (bit) and connection method; printer — dpi, printing speed (ppm) and connection method; monitor — size, resolution and pixel density (ppi); projector — resolution, brightness (lm) and connection method.

Touch screen: intuitive and saves space; suitable for simple selections. Keyboard and mouse: faster and more precise for text entry; suitable for long typing sessions.

Ergonomics: avoid glare from curtains or lighting, maintain even brightness, adjust chair and desk height, support the wrists and take regular breaks.
*Common mistakes: writing only ‘convenient’ or ‘easy to use’; comparing without a common criterion; treating a printer as an input device; confusing resolution (dpi) with printing speed (ppm).

3. Similar Questions #Same angle, different device

1. The library's book search system used a keyboard and mouse, but now uses a touch screen.

(a) State **two** benefits of using a touch screen.

【Answer】Intuitive operation; readers do not need to learn to use a mouse; saves space because the screen can be embedded; suitable for simple selections (buttons, zooming in and out).

(b) State one way a touch screen is less effective than a keyboard and mouse.

【Answer】Slower text entry; more tiring during long input; lower accuracy (easier to press the wrong item); fingerprints on the screen.

2. The school hall will install a projector to show teachers' presentations and videos.

(a) Explain why projector brightness (lm) is an important specification.

【Answer】The hall is large and has more ambient light; insufficient brightness makes the image dim and colours unclear, so students at the back cannot see clearly.

(b) Apart from brightness, state one projector specification to consider and explain it.

【Answer】Resolution: higher resolution gives a sharper image and clearer text; connection method: HDMI/Wi-Fi must match the existing computer; contrast ratio: better detail in dark areas.

3. Mr Yip types at a computer for six hours a day and has recently had wrist and neck discomfort.

(a) State **two** settings or practices that can reduce health problems.

【Answer】Use an ergonomic keyboard/wrist rest; adjust the monitor to eye level; use a chair with back support and keep feet flat; take a break and stretch every 30–60 minutes.

(b) Explain why a monitor positioned too low can cause neck discomfort.

【Answer】A monitor that is too low requires the user to look down; the neck stays bent forward for a long time and the muscles remain tense, leading to fatigue and pain.